

Hydro-Dynamic Atomization and Ultrasonic Water Mist System for Dust Suppression in Continuous Mining Operations

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1. Abstract

Respirable coal dust generated during continuous miner shearer operations presents severe occupational safety and pneumoc... This paper evaluates a high-pressure hydro-dynamic atomization curtain operating at 45 L/min water delivery rate. Field experiments demonstrate an overall respirable dust reduction efficiency of 86.4% compared to conventional low-pressure

Keywords: Dust suppression, Respirable coal dust, Hydro-dynamic atomization, Underground mining

2. Introduction

Continuous mining shearers generate fine airborne dust particles below 10 micrometers. Conventional water sprays fail to capture sub-micron dust due to high droplet surface tension..

3. Methodology

We utilized Computational Fluid Dynamic (CFD) multi-phase spray modeling combined with Response Surface Methodology (RSM). Water pressure was varied from 3.5 MPa to 8.0 MPa..

4. Results

At 6.5 MPa operational pressure, respirable dust concentration at the return airway dropped from 14.2 mg/m³ to 1.93 mg/m³. The system achieved a total dust reduction of 86.4% and an F1-score of 0.885 for automated spray activation..